

HAO ZHEN

706-765-5581 | hz62826@uga.edu | [Google Scholar](#) | Athens, GA

SUMMARY

Traffic safety researcher and engineer with expertise in AI/ML applications for transportation, crash analysis and predictive modeling, connected autonomous vehicles, and transportation data analytics. Experience leading research projects funded by GDOT, FHWA, USDOT, and NSF. Proficient in Python, deep learning frameworks, statistical analysis, and GIS tools with 16+ peer-reviewed publications.

TECHNICAL SKILLS

Programming & Software: Python, R, SQL, MATLAB, PyTorch, TensorFlow, Keras, Scikit-learn, ArcGIS, QGIS
Transportation Tools: Synchro, VISSIM, HCS, Highway Safety Manual Methods, MOVES
Data Science: Machine Learning, Deep Learning, Graph Neural Networks, Large Language Models, Computer Vision, Natural Language Processing, Statistical Modeling, Time Series Forecasting
Analysis Methods: Crash Prediction Modeling, Safety Performance Functions, Network Optimization, Spatial Analysis, Before-After Studies, Regression Analysis, Classification Methods

EDUCATION

- University of Georgia** 2021 – 2025
Ph.D. in Civil and Environmental Engineering Athens, GA
 - Dissertation: *Bridging Structure and Semantics: A Multimodal Interpretable LLM Framework for Traffic Crash Modeling and Analysis*
 - Advisor: Dr. Jidong J. Yang
- North China Electric Power University** 2018 – 2021
M.S. in Technology Economics and Management Beijing, China
 - Thesis: *Multi-source Complementary Micro-grid Energy Intelligent Prediction and Optimized Management*
- North China Electric Power University** 2014 – 2018
B.E. in Renewable Energy Science and Engineering Beijing, China

PROFESSIONAL EXPERIENCE

- University of Georgia** October 2025 – Present
Postdoctoral Researcher Athens, GA
 - Developing AI-powered multimodal sensing and decision-making tools for rural incident management
 - Leading research on vision-language models for automated crash diagram generation
 - Collaborating with GDOT on roundabout safety analysis and freeway shoulder width studies
- University of Georgia** 2021 – 2025
Graduate Research Assistant Athens, GA
 - Led traffic crash modeling research using machine learning, deep learning, and large language models
 - Developed graph neural network models for statewide AADT estimation integrating network topology
 - Created optimization framework for continuous count station and WIM site placement across Georgia
 - Conducted crash severity analysis bridging tabular crash data with contextual natural language information
 - Analyzed healthcare accessibility and transportation equity using mobile device mobility data
- University of Georgia** 2024 – 2025
Graduate Teaching Assistant Athens, GA
 - Courses: Advanced Deep Learning, Statistical and Machine Learning, AI in Civil Engineering
 - Developed coding labs and Jupyter notebooks demonstrating ML applications in transportation

SELECTED PROJECTS

- Assessing Safety Effects of Lighting at Roundabouts in Georgia** 2025–Present
Georgia Department of Transportation (GDOT)
 - Analyzing crash data and geometric features to evaluate safety performance at roundabout locations
 - Developing crash modification factors for lighting treatments using before-after studies
- Safety Effectiveness of Inside Shoulder Widths on Freeways** 2024–2025
Georgia Department of Transportation (GDOT)
 - Conducted statistical analysis of crash frequency and severity related to shoulder width variations
 - Applied Highway Safety Manual methods to quantify safety impacts on Georgia freeways
- Flood-Induced Transportation Network Disruption** 2024–2025
Fort Stewart/Hunter Army Airfield
 - Modeling network vulnerability and resilience under flood scenarios for military installation access
- Leveraging Probe Data for Incident Management in Rural Areas** 2024
FHWA & GDOT
 - Developed data-driven approaches using probe vehicle data to improve rural incident detection
 - Created visualization tools for real-time incident monitoring and response optimization
- Strategic Planning of Continuous Count Stations and WIM Sites** 2021–2023
Georgia Department of Transportation (GDOT)
 - Created optimization framework for statewide sensor network placement using clustering algorithms
 - Applied information gain metrics to prioritize monitoring locations for traffic data collection
 - Delivered interactive visualization tools and recommendations to GDOT Office of Transportation Data

• Community-Oriented Mobility Modeling for Rural Communities <i>USDOT Region 4 UTC</i> ◦ Analyzed healthcare accessibility patterns across primary, pediatric, urgent, and emergency care ◦ Investigated transportation equity at census block group level using mobile mobility data • Eco-driving Trajectory Planning for Heterogeneous Platoons <i>National Science Foundation (NSF)</i> ◦ Developed trajectory optimization algorithms for connected autonomous vehicle platoons in urban environments ◦ Implemented stochastic coordination methods for mixed human-driven and automated vehicle platoons	2023–2025 2021–2022
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SELECTED PUBLICATIONS

(16+ TOTAL PUBLICATIONS)

1. **Zhen, H.**, Yang, J.J. (2025). CrashSage: A Large Language Model-Centered Framework for Contextual and Interpretable Traffic Crash Analysis. *Artificial Intelligence for Transportation*.
2. **Zhen, H.**, Yang, J.J. (2025). Tab-text: Bridging Tabular Data and Natural Language for Enhanced Traffic Safety Analysis and Modeling. *Expert Systems with Applications*.
3. **Zhen, H.**, Yang, J.J. (2024). Analyzing the Importance of Network Topology in AADT Estimation: Insights from Travel Demand Models using Graph Neural Networks. *Transportation*.
4. **Zhen, H.**, et al. (2024). Leveraging Large Language Models with Chain-of-Thought and Prompt Engineering for Traffic Crash Severity Analysis. *Computers*.
5. **Zhen, H.**, et al. (2024). Unraveling the Dynamics of Single-vehicle versus Multi-vehicle Crashes: A Comparative Analysis through Binary Classification. *Applied Computing and Intelligence*.
6. **Zhen, H.**, et al. (2023). Co-supervised Learning Paradigm with Conditional GANs for Sample-efficient Classification. *Applied Computing and Intelligence*.
7. **Zhen, H.**, et al. (2022). Eco-driving Trajectory Planning of a Heterogeneous Platoon in Urban Environments. *IFAC AAC 2022*.
8. Huang, Y., **Zhen, H.**, Yang, J.J. (2024). Cluster-Guided Denoising Graph Auto-Encoder for Traffic Data Imputation. *Expert Systems with Applications*.
9. Lares, O., **Zhen, H.**, Yang, J.J. (2025). Feature Group Tabular Transformer: Traffic Crash Modeling and Causality Analysis. *Applied Computing and Intelligence*.

SELECTED PRESENTATIONS

Podium: TRB 105th Annual Meeting (2026), ASCE ICTD Atlanta (2024), IFAC AAC Columbus (2022)
Poster: TRB Annual Meetings (2023–2026), GDOT/GTI Research Expo (2022, 2024), AI in Motion Symposium Austin (2024), Joint CATM and CR2C2 Symposium (2024)

HONORS AND AWARDS

• Certificate of Appreciation <i>USDOT Region 4 University Transportation Center (CR2C2)</i> • Injury Control and Emergency Health Services Student Scholarship <i>American Public Health Association</i> • Helene M. Overly Memorial Scholarship <i>Women's Transportation Seminar (WTS)</i> • Scholarship for Outstanding Performance <i>North China Electric Power University</i>	2025 2025 2022 2015–2020
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PROFESSIONAL AFFILIATIONS & SERVICE

Committees: ASCE Artificial Intelligence in Transportation Committee (Corresponding Member), ASCE Transportation Safety Committee (Corresponding Member), American Planning Association (Student Member), PAKDD 2026 Program Committee
Leadership: UGA Student Leader for CR2C2 USDOT Region 4 UTC; Organized research seminars and K-12 outreach; Judge for ASCE Southeast Student Symposium Traffic Control Competition (2025)
Reviewer: Transportation Research Board Annual Meeting, IEEE Transactions on Vehicular Technology, Journal of Cleaner Production, Energy, Neural Computing & Applications, Environmental Science and Pollution Research